Neuro-Adaptive Ensemble Framework for EEG-Based Emotion Recognition

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***Abstract*—Emotion recognition using physiological signals is a rapidly expanding area in affective computing. While single-classifier machine learning models have shown promise in classifying neural states, they often struggle with the non-linear and non-stationary nature of electroencephalography (EEG) signals. This paper presents a Neuro-Adaptive Ensemble Framework designed to classify emotional states across four dimensions: Valence, Arousal, Dominance, and Liking, using the DEAP dataset. Our approach decomposes 3-second EEG epochs using Empirical Mode Decomposition (EMD) to capture intrinsic oscillatory dynamics. We extract 52 features spanning temporal, spectral, and wavelet domains from both the raw signals and the extracted Intrinsic Mode Functions (IMFs). To combat high dimensionality, we introduce a three-stage feature selection pipeline (correlation filtering, information gain ranking, and recursive feature elimination) to isolate the most discriminative attributes. Finally, a soft-voting ensemble combining Random Forest, XGBoost, and an SVM estimator classifies the selected features. Evaluated via 5-fold stratified cross-validation, the proposed framework achieves accuracies of 76.82%, 75.13%, 76.01%, and 77.34% across the four dimensions, consistently outperforming standalone baseline models.**

***Index Terms*—Affective Computing, EEG, Empirical Mode Decomposition, Ensemble Learning, Feature Selection, DEAP.**

I. Introduction

Human emotion plays a fundamental role in cognition, social interaction, and decision-making. In recent years, training computational systems to recognize and adapt to these emotional states—a field known as affective computing—has gained significant traction. Among the various physiological markers used to infer emotion, Electroencephalography (EEG) is highly valued as it captures neuro-electrical activity directly from the central nervous system, offering a direct window into affective states.

However, extracting reliable emotional indicators from EEG data is notoriously difficult. The signals are inherently non-stationary, highly susceptible to noise, and vary drastically across different subjects. Previous studies have heavily relied on single classifiers to interpret these signals. For instance, Parui et al. demonstrated that using an Extreme Gradient Boosting (XGBoost) algorithm following a correlation and recursive feature elimination pipeline yielded strong classification results on the DEAP dataset. Yet, relying on a single learning algorithm can introduce high prediction variance when faced with heterogeneous neural patterns. Furthermore, standard feature extraction techniques often fail to fully exploit the adaptive, dynamic nature of brainwaves.

To address these limitations, this study proposes a comprehensive, multi-stage pipeline termed the Neuro-Adaptive Ensemble Framework. We introduce two primary methodological upgrades:

1) The integration of Empirical Mode Decomposition (EMD) as a vital pre-processing step, separating complex EEG signals into simpler Intrinsic Mode Functions (IMFs) without spectral leakage.
2) A soft-voting ensemble mechanism that mitigates individual model bias by combining the predictive power of Random Forest, XGBoost, and Support Vector Machines (SVM).

II. Methodology

The proposed system architecture is designed as a sequential, modular pipeline consisting of data preparation, signal decomposition, feature engineering, and ensemble classification.

A. Dataset and Pre-processing

This research utilizes the publicly available DEAP (Database for Emotion Analysis using Physiological Signals) dataset. The dataset comprises 32-channel EEG recordings from 32 participants who were exposed to 40 music video clips, each lasting 60 seconds. Following the stimuli, participants provided self-assessments on a continuous 1 to 9 scale across four dimensions: Valence, Arousal, Dominance, and Liking.

For our binary classification task, these continuous ratings were thresholded at 4.5. The pre-processed EEG data (down-sampled to 128 Hz with baseline removal and ICA-based artifact rejection applied) was segmented into non-overlapping 3-second windows, resulting in 20 epochs per trial and a total corpus of 25,600 labeled epochs.

B. Empirical Mode Decomposition (EMD)

EEG signals lack a fixed frequency basis. EMD is an adaptive, data-driven technique that breaks down non-linear signals into a series of Intrinsic Mode Functions (IMFs) and a residual trend. We applied standard EMD to every 3-second epoch across all 32 channels. To optimize computational efficiency while capturing the highest-frequency oscillatory components most relevant to emotional shifts, only the top three IMFs (IMF0, IMF1, and IMF2) were retained. These were concatenated with the raw signal, effectively quadrupling the richness of our signal representation.

C. Feature Extraction

To construct a robust representation of the neural data, a comprehensive suite of 52 features was computed per expanded signal vector:

- **Time-Domain (8):** Coefficient of variation, Higuchi fractal dimension, kurtosis, skewness, and differential statistics.
- **Hjorth Parameters (3):** Activity, Mobility, and Complexity.
- **Autoregressive Parameters (9):** Derived using a Burg AR model of order 4.
- **Frequency Spectral (12):** FFT-based absolute powers across standard bands (Delta, Theta, Alpha, Beta, Gamma) and targeted spectral ratios.
- **Wavelet-Domain (20):** Statistics derived from a 4-level Daubechies-4 Discrete Wavelet Transform.

D. Cascaded Feature Selection

Extracting features across multiple IMFs and channels results in a high-dimensional space prone to redundancy. We utilized a three-stage dimensionality reduction pipeline tailored independently for each emotion scale:

1) **Correlation Removal:** Pairwise correlation was calculated, and redundant attributes with an absolute Pearson coefficient $|r| \geq 0.90$ were discarded.
2) **Information Gain Ranking:** A Random Forest estimator calculated the Mean Decrease in Impurity (MDI) to rank the surviving features.
3) **Recursive Feature Elimination (RFE):** Utilizing a 5-fold stratified cross-validation approach, the weakest features were iteratively pruned until optimal performance was isolated.

E. Ensemble Classification

To classify the final selected features, we deployed a soft-voting meta-learner. This ensemble aggregates the posterior class probabilities from three diverse base estimators:

- **Random Forest:** Provides variance reduction through bootstrap aggregation (200 trees).
- **XGBoost:** Reduces bias through gradient boosting (200 estimators, learning rate = 0.05, max depth = 6).
- **RBF-SVM:** Offers margin-based classification in a transformed high-dimensional space.

The meta-learner assigns weights proportional to each classifier's independent validation accuracy, heavily leveraging the XGBoost predictions while stabilizing them with the RF and SVM outputs.

III. Experimental Results

The framework was evaluated using a 5-fold stratified cross-validation protocol to ensure rigorous testing and prevent data leakage.

A. Feature Selection Efficacy

The cascaded selection process aggressively reduced the feature space, optimizing computational load without sacrificing predictive power. Table I illustrates the feature reduction at each stage.

TABLE I
FEATURE COUNT PROGRESSION PER EMOTION SCALE

Emotion	Initial	Post-Corr.	Post-IG	Post-RFE
Valence	52	43	7	7
Arousal	52	43	27	16
Dominance	52	43	20	18
Liking	52	43	15	11

B. Classification Performance

We benchmarked our Neuro-Adaptive Ensemble against several standard classifiers, as well as the standalone XGBoost baseline model documented in prior literature for this specific dataset.

TABLE II
CLASSIFICATION ACCURACY COMPARISON

Classifier	Valence	Arousal	Dominance	Liking
Naive Bayes	72.19%	71.28%	73.28%	74.22%
Decision Tree	72.47%	72.00%	73.71%	74.56%
Random Forest	74.03%	73.56%	74.47%	75.87%
XGBoost (Baseline)	75.97%	74.20%	75.23%	76.42%
Proposed Ensemble	**76.82%**	**75.13%**	**76.01%**	**77.34%**

As detailed in Table II, the proposed ensemble framework achieved the highest accuracy across all four affective dimensions. The inclusion of EMD processing combined with ensemble learning provided a consistent absolute improvement of 0.78% to 0.93% over the standalone XGBoost baseline. While modest, this gain represents a statistically consistent reduction in classification variance across subjects.

IV. Conclusion

This study validates the efficacy of combining adaptive signal decomposition with ensemble machine learning for EEG-based emotion recognition. By integrating Empirical Mode Decomposition to extract intrinsic oscillatory modes, extracting a rich 52-feature matrix, and applying a rigorous selection pipeline, the framework successfully isolates critical neural indicators of emotion. The soft-voting ensemble effectively balances bias and variance, resulting in superior classification accuracy compared to traditional single-model approaches. Future work will focus on evaluating the framework under a Leave-One-Subject-Out (LOSO) cross-validation scheme and exploring the fusion of peripheral physiological signals to further enhance predictive robustness.

References

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